Hypertension

REVIEW

Heritability and Cardiovascular Disease in Adult Offspring of Hypertensive Disorder of Pregnancy: A Systematic Review and Meta-Analysis

Ansu Mari Saji, Reeya Chandarana, Tiberiu A. Pana, Ben Carter, Phyo Kyaw Myint

ABSTRACT: The heritability of hypertensive disorders of pregnancy (HDP) and HDP's association with cardiovascular disease in adult offspring were explored through a systematic review and meta-analysis. MEDLINE and EMBASE were searched independently by 2 reviewers (inception to February 18, 2025). Maternal chronic hypertension, antenatal complications, and pediatric cardiovascular disease were excluded. The Critical Appraisal Skills Program checklist was used for critical appraisal. Random-effects meta-analysis was conducted using a generic inverse variance method. Narrative synthesis and pooled results were expressed as odds ratios with 95% CIs. Of 225 studies screened, 11 studies (n=59 185; 48.3% women) assessed cardiovascular disease, and 9 studies (n=58 512) assessed HDP. Offspring age ranged from 19 to 55 years. There were 11 good- and 9 fair-quality studies. Fifteen studies were included in the meta-analysis (n=266 244). Adult offspring exposed to HDP had systolic and diastolic blood pressure that was 3.40 mm Hg (95% CI, 2.44–4.37; I^2=40%) and 2.19 mm Hg higher (95% CI, 1.40–2.98; I^2=51%). Higher odds of hypertension were observed (odds ratio, 1.50 [95% CI, 1.18–1.91]; I^2=66%). Female offspring exhibited 72% increased odds of HDP (OR, 1.72 [95% CI, 1.41–2.09]; I^2=81%) and 90% increased odds of preeclampsia (odds ratio, 1.90 [95% CI, 1.47–2.46]; I^2=36%) following exposure to the same disorders. Increased odds of premature acute coronary syndrome and higher stroke risk were observed following HDP and severe preeclampsia exposure, respectively. In adult offspring, heritability of HDP was high. HDP were associated with hypertension, acute coronary syndrome, and stroke. Lifestyle modifications, cardiovascular disease monitoring, and prevention are paramount.

Key Words: adult ■ blood pressure ■ child ■ middle aged ■ stroke

Hypertensive disorders of pregnancy (HDP) affect ≈5% to 10% of pregnancies globally. Preeclampsia, gestational hypertension, superimposed preeclampsia, and chronic hypertension are leading causes of maternal and perinatal morbidity and mortality.[1–3] Preeclampsia is a long-term risk factor for maternal cardiovascular disease, including a 4-fold increased risk of heart failure and a 2-fold increased risk of coronary heart disease and stroke.[4] Intergenerational effects are also seen: daughters of preeclamptic pregnancies are at more than double the risk of experiencing preeclampsia themselves.[5,6] In a UK cohort of 17 302 participants, the risk of preeclampsia in daughters whose mothers had preeclampsia was nearly 3-fold. Similar findings were observed in Israeli and Japanese cohorts, suggesting a familial predisposition to HDP and cardiovascular risk.[7,8]

The effect of HDP exposure on offspring cardiovascular health, specifically in adulthood, is poorly understood. Preeclampsia and gestational hypertension were linked to a pooled increase of 2.70 mm Hg and 1.39 mm Hg higher systolic blood pressure (SBP) and diastolic BP (DBP), respectively.[9] However, only 25% of included studies followed offspring to adulthood. At 20 years of age, offspring exposed to HDP were 2.5× more likely to have a lifetime QRISK score above the 75th centile, indicating a ≥40% chance of developing cardiovascular disease. Consistent with these findings, preeclampsia and gestational hypertension were associated with stroke but not coronary heart disease in a Finnish cohort of 6410 participants.[10] Conversely, an Indonesian population-based case-control study of 306 participants found that pregnancy-related high

Correspondence to: Phyo Kyaw Myint, University of Aberdeen, Room 4.013, Polwarth Bldg, Foresterhill, Aberdeen, AB25 2ZD, Scotland, United Kingdom. Email phyo.myint@abdn.ac.uk

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Nonstandard Abbreviations and Acronyms

ACS	acute coronary syndrome
BP	blood pressure
CVD	Cardiovascular disease
DBP	diastolic blood pressure
FMF	Fetal Medicine Foundation
HDP	hypertensive disorders of pregnancy
OR	odds ratio
PAPP-A	pregnancy-associated plasma protein A
PlGF	protein placental growth factor
RR	relative risk
SBP	systolic blood pressure

BP was associated with premature acute coronary syndrome.[11]

Although HDP exposure is associated with hypertension in childhood, it is uncertain whether this confers an increased risk of cardiovascular events in adulthood, particularly at middle to older ages. Furthermore, the risk of HDP in daughters affected by HDP has not been systematically reviewed. The aim of this meta-analysis was (1) to investigate cardiovascular disease risk in adult offspring born to HDP compared with those of normotensive pregnancies and (2) to determine the heritability of HDP among mothers and daughters.

METHODS

Data Availability

All data relevant to the study are included in the article or uploaded as supplementary information.

Protocol and Registration

This review was performed according to the PRISMA checklist (Tables S1 and S2).[12] The protocol was registered with PROSPERO (REGISTRATION: URL: https://www.crd.york.ac.uk/PROSPERO/; Unique identifier: CRD42021265145) on July 2, 2021.

Search Strategy

A systematic review search was conducted on February 18, 2025, using MEDLINE and EMBASE. The search was limited to studies involving humans, with no date or language restrictions applied. Reference lists of included studies were reviewed to identify additional studies. The search involved keywords and medical subject headings (MeSH) for synonyms of "hypertensive disorders of pregnancy" and "adult offspring." The "explode" function was used on the following: "cardiovascular diseases," "cardiomyopathies," "heart disease risk factors," and "myocardial infarction." Truncations using Boolean operators (AND/OR) were also utilized. The full search strategy is outlined in Table S3.

Eligibility Criteria

Studies assessing HDP compared with normotensive pregnancies in women with live infants were selected. Cardiovascular outcomes in adult offspring aged 18 years and above were evaluated between the 2 groups. Outcomes included hypertension, myocardial infarction, heart failure, and stroke. The presence of HDP in adult daughters exposed to HDP versus normotensive pregnancies was also assessed. Observational studies (eg, case-control, cohort, cross-sectional) were selected.

The following were excluded: antenatal complications (eg, antepartum hemorrhage, miscarriages), chronic hypertension, neonatal deaths, stillbirths, pediatric cardiovascular disease, and congenital heart disease. Randomized controlled trials, editorials, and studies deficient in primary data were also excluded.

Screening and Data Extraction

All studies were exported to Rayyan review software.[13] Duplicates were removed, and titles, abstracts and full texts were independently screened by 2 reviewers (AMS and RC). Any disagreements were discussed and arbitrated by a third reviewer (SB). Data were extracted independently by 2 reviewers (AMS and RC) and cross-checked for discrepancies. Microsoft Excel was used for data extraction. The following were obtained: study and subject characteristics, definition of HDP and control groups, outcomes, and confounders (Table S4).

Risk of Bias Assessment

The Critical Appraisal Skills program checklist for quality assessment of cohort and case-control studies was utilized.[14] Two reviewers independently appraised each study, and inconsistencies were resolved via discussion. Studies were assessed to be at low, moderate, or high risk of bias, corresponding to good, fair, or poor quality. Critical Appraisal Skills program assessments of "No," "Unsure," and "Yes" were given the scores 1, 2, and 3, respectively. For section A: sum of scores <12, 12 to 18 and >18 were interpreted as poor, fair and good. Similarly, for sections B and C: sum of scores from 0 to 3, 4 to 6, and 7 to 9 were interpreted as poor, fair, and good. Studies were deemed to be of overall good quality when all sections scored good.

The certainty of evidence was assessed by 2 reviewers (AMS, RC) using the Grading of Recommendations, Assessment, Development and Evaluation approach and GRADEpro GDT.[15,16] Outcomes from observational studies were deemed to be of low certainty. Certainty was downgraded to very low following limitations in risk of bias, inconsistency, indirectness, imprecision or publication bias. Certainty was upgraded when the magnitude of effect was assessed as large or very large or when confounding was expected to decrease the observed effect or increase the effect where no effect was seen (Tables S5 and S6).

Outcomes

The primary outcomes were cardiovascular disease and HDP in adult offspring. Cardiovascular disease, including hypertension, coronary artery disease, premature acute coronary syndrome, and stroke was measured using adjusted and unadjusted relative risks (RRs), odds ratios (ORs) or hazard ratios. SBP and DBP were measured as mean or mean difference. HDP in

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female adult offspring was reported as RR or OR. Unadjusted and adjusted values (adjusted for a minimum of offspring sex for hypertension and offspring age for HDP) were extracted.

Data Synthesis

SBP and DBP in adult offspring exposed to HDP versus unexposed were pooled as mean differences with 95% CIs. All papers reported mean BP and SD. Cochrane Collaboration software RevMan (Version 5.4.1) was used to calculate the mean difference in BP (Table S7).[17]

Hypertension and HDP in adult offspring were pooled as ORs with 95% CIs. The generic inverse variance was used as the weighting alongside the DerSimonian-Laird estimator method for pooling random effects. Due to clinical and methodological variance within included studies, a random-effects model was used. Log (odds ratio) was calculated using the ln function and SE using the equation (SE=(ln(upper limit)−ln(odds ratio))/1.96).[18] Raw data in Tables S8 through S14.

It was recognized that not all studies would adjust for the same confounders. Therefore, the most important factor was identified (offspring sex for hypertension and offspring age for HDP). This minimum common adjusting variable was necessary for studies to be considered adjusted. Studies ineligible for statistical pooling were reported as a narrative synthesis.

Heterogeneity and Subgroup Analysis

Statistical heterogeneity was assessed using the I^2 statistic. I^2 of 0%, 25%, 50%, and 75% were interpreted as no, low, moderate, and high levels of heterogeneity.[19] Preagreed subgroups (eg, HDP subtypes) were used to explore heterogeneity. Funnel plots were used to evaluate publication bias when ≥10 studies were pooled (Figure S1). To assess the effect of excluding maternal chronic hypertension and preterm offspring on offspring BP, sensitivity analysis was conducted.

RESULTS

Study Selection

The study selection process is illustrated in the PRISMA flowchart (Figure 1). A total of 271 studies with 46 duplicates were identified. Following duplicate removal, 225 studies underwent title, abstract, and full-text screening. A further 8 studies were identified by reference searching. A total of 20 studies were eligible for inclusion. Nine studies were excluded: 1 study encompassed the incorrect population (children),[20] 3 studies investigated different outcomes,[21–23] and 5 studies were of excluded on the basis of study design (ie, abstract,[24–26] review,[27] and letter[28]).

Study Characteristics

Characteristics of included studies are summarized in the Table and Tables S20 and S21. Risk estimates reported and confounders adjusted for in each study are summarized in Tables S15 through S19.

Among the 20 observational studies that met the eligibility criteria, 11 studies examined cardiovascular outcomes, and 9 studies investigated HDP in adult offspring with a maternal history of HDP.

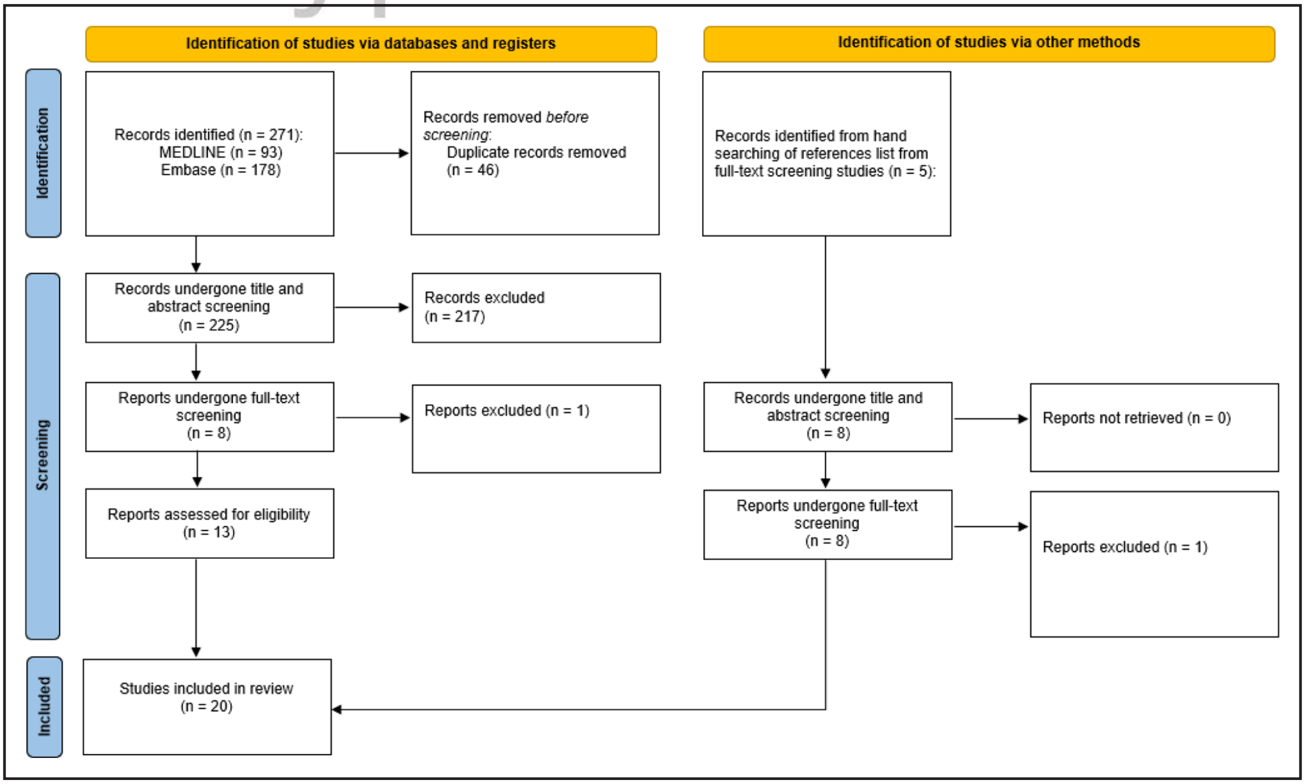

Figure 1. PRISMA 2020 flowchart for new systematic reviews that included searches of databases, registers, and other sources.

Table. Characteristics of Studies Assessing Cardiovascular Disease in Adult Offspring Exposed to HDP

Study	Country of origin	Study design	Mean maternal age in years (SD)		Offspring age at follow-up (years)	Population	Female (%)
			HDP	No HDP			
Alsnes et al 2017[36]	Norway	Prospective CS	26.7 (6.0)	25.7 (5.4)	No hypertension: 28.9 (6.2)* Any hypertension: 28.4 (6.1)*	15 778	55.7
Davis et al 2015[29]	Australia	Prospective CS	29.1 (5.8)	29.4 (5.7)	20	1151	46.0
Dines et al 2023[35]	United States	Retrospective CS	26.7 (5.5)	26.9 (4.6)	31.8 (19.3–38.8)†	8755	48.2
Kajantie et al 2009[10]	Finland	Retrospective CS	Gestational hypertension: 28.9 (5.8) Non severe preeclampsia: 27.7 (5.0) Severe preeclampsia: 29.1 (5.9)	27.8 (5.2)	30.0 (2.9)*	6147	NA
Kurbasic et al 2018[32]	Sweden	Retrospective CS	NA	NA	40	13 893	49.4
Lazdam et al 2010[37]	United Kingdom	Prospective CS	NA	NA	20–30 y‡	109	51.4
Mamun et al 2011	Australia	Prospective CS	NA	NA	21	2608	50.4
Mogren et al 2001[33]	Sweden	Retrospective CS	NA	NA	29–41	7876	50.1
Palmsten et al 2010[34]	United States	Prospective CS	Younger than 20–20.4% 20–35 y, 68.4% 36 or older, 11.2%	Younger than 20–17.8% 20–35 y, 76.4% 36 or older, 5.8%	39† (range 34–44 y)	1556	59.3
Qanitha et al 2017[11]	Indonesia	Case-Control	NA	NA	47 (6.3)* (range 28–55 y)	306	18.3
Tapp et al 2018[31]	Finland	Prospective CS	NA	NA	40* (range 34–49 y)	1006	50.9

CS indicates cohort study; HDP, hypertensive disorders of pregnancy; and NA, not available.
*Mean (SD).
†Median (interquartile range).
‡Range.

A total of 59 185 participants were included in studies assessing cardiovascular outcomes (Table). The age of offspring at follow-up ranged between 19 and 55 years. Mean maternal age was higher in mothers without HDP in Davis et al 2015, Dines et al 2023, and Kajantie et al 2009 (nonsevere preeclampsia only). Male and female offspring were in equal proportion in most studies, except Qanitha et al 2017. The percentage of female offspring reported was 18.3%. Two studies originated from Australia,[29,30] 2 from Finland,[10,31] 2 from Sweden,[32,33] 2 from the United States,[34,35] 1 from Indonesia,[11] Norway, and the United Kingdom.[36,37] There were 5 retrospective and 5 prospective cohort studies and 1 case-control study.[10]

A total of 58 512 participants were included across the 9 retrospective cohort studies investigating HDP in adult offspring. Two papers originated from the United Kingdom,[6,38] 2 from Sweden,[39,40] 1 from Iceland,[41] Israel,[7] Japan,[8] Norway[5] and the United States[42] (Tables S20 and S21). Median age at first pregnancy was described by 1 study and was higher in those exposed to HDP.[5]

Among the 20 studies, controls were normotensive pregnancies, except for the following: Lazdam et al 2010, Chesley et al 1986, and Sutherland et al 1981. Control groups for these studies included: age-matched uncomplicated pregnancies, another pregnancy, and matched normotensive mother-in-law, respectively.

HDP definitions and measurements used in each study are summarized in Tables S22 and S23. In the studies assessing cardiovascular risk, 6 studies used similar definitions of HDP. Gestational hypertension was defined as BP ≥140/90 mm Hg after 20 weeks of gestation and preeclampsia as the former with proteinuria (≥0.3 g/24 hours or ≥2+ on dipstick).[10,29–31,37] Kajantie et al 2009 further divided preeclampsia as severe (BP ≥160/110 mm Hg) and nonsevere (BP ≥140/90 mm Hg). Davis et al 2015 used the term pregnancy-induced hypertension to describe gestational hypertension and complicated pregnancy hypertension for preeclampsia. Palmsten et al 2010 defined HDP as any of the following after 24 weeks of gestation: increase of SBP (≥ 30 mm Hg) or DBP ≥ (15 mm Hg) above usual on 2 occasions 6 hours apart or proteinuria (≥+1 on dipstick, >30 mg) on ≥2 days in the absence of a urinary tract infection or persistent edema of the hands and face. Tapp et al 2018 described HDP as hypertension only during pregnancy.

Alsnes et al 2017 included 27 participants with maternal chronic hypertension in the normotensive group. In addition, Lazdam et al 2010 reported SBP and DBP in hypertensive pregnancies of preterm offspring compared with normotensive pregnancies. Subgroup analyses excluding Alsnes et al 2017 and Lazdam et al 2010 were performed.

In the papers analyzing the heritability of HDP among daughters, 4 studies defined gestational hypertension as BP ≥140/90 mm Hg after 20 weeks of gestation and preeclampsia as the former with proteinuria (≥0.3 g/24 hours).[7,8,39,40] Ayorinde et al 2017 used a lower BP threshold at ≥110/90 mm Hg. Skjærven et al 2005 referred to bmj.com for definitions. Furthermore, the Nelson classification 1965 and 1955a, respectively, were utilized by 2 studies.[38,42] The 1955a classification outlined mild preeclampsia as DBP ≥90 mm Hg after 26 weeks of gestation on 2 occasions at least 1 day apart or progressive increase during labor. Severe preeclampsia was described as proteinuria exceeding 0.25 g/L. Arngrimsson et al, 1990 further defined severe preeclampsia as BP ≥160/110 mm Hg with or without proteinuria and eclampsia as seizures in hypertensive pregnancy after 20 weeks of gestation or within 24 hours of delivery.

Two studies analyzed HDP subtypes separately. Ayorinde et al, 2017 investigated gestational hypertension in those exposed to gestational hypertension and preeclampsia, and vice versa. Meanwhile, Nilsson et al, 2004 assessed preeclampsia in those exposed to preeclampsia and gestational hypertension. Odds of gestational hypertension in those exposed to maternal gestational hypertension were also assessed. Moreover, 2 studies investigated risk ratios according to birth order. Mogren et al 1999 reported the RR of preeclampsia in elder daughters' first and second deliveries and younger daughters' first and second deliveries separately. Similarly, Skjærven et al, 2005 reported the odds of preeclampsia in the daughter's first and subsequent pregnancies.

Assessment of Study Quality

Critical Appraisal Skills program checklist for quality assessment is summarized in Tables S24 and S25. For studies assessing cardiovascular disease in adult offspring, 9 studies were of good quality and 2 studies of fair quality. For studies assessing the heritability of HDP in daughters, 7 studies were of fair quality and 2 studies of good quality.[43]

Primary Outcome: Cardiovascular Disease Risk

Of the 11 studies that investigated cardiovascular outcomes: 6 assessed the relationship between HDP and mean difference in SBP and DBP.[29–32,36,37] Higher BP was reported in offspring exposed to HDP in all studies, except Lazdam et al 2010, which reported higher SBP in offspring unexposed to HDP. Five studies reported positive associations between HDP and hypertension,[10,29,32–35] 2 between HDP and coronary heart disease,[10,11] and 1 between HDP and stroke.[10]

Six studies (n=34, 507) analyzed the relationship between HDP and mean difference in blood pressure.[29–32,36,37] All studies reported mean blood pressure and SD. Following meta-analysis, HDP was associated with a higher mean SBP difference (3.40 mm Hg [95% CI, 2.44–4.37], I^2=40%, P<0.00001; Figure 2) and DBP difference (2.19 mm Hg [95% CI, 1.40–2.98], I^2=51%, P<0.00001; Figure 3).

Subgroup analysis, excluding maternal chronic hypertension, found higher SBP (3.79 mm Hg [95% CI, 2.68–4.89], I^2=22%) and DBP (2.58 mm Hg [95% CI, 1.84–3.32], I^2=11%; Figures S2 and S3).

HDP and Hypertension

Five studies (n=29, 988) analyzed the relationship between HDP and hypertension.[10,29,32–35] Four studies reported hypertension using unadjusted ORs and were pooled (OR, 1.67 [1.41–1.99], I^2=12%, P<0.0001; Figure S6). Similarly, 4 studies reported adjusted ORs. The pooled adjusted OR showed 50% increased odds of hypertension in adult offspring exposed to HDP (OR, 1.50 [95% CI, 1.18–1.91], I^2=66%, P=0.001; Figure 4). A subgroup analysis assessing the effect of HDP subtype—preeclampsia—was performed (Figure S7). A nonsignificant increase in odds of hypertension (OR, 1.70 [95% CI, 0.79–3.69], I^2=79%; P=0.18) was observed.

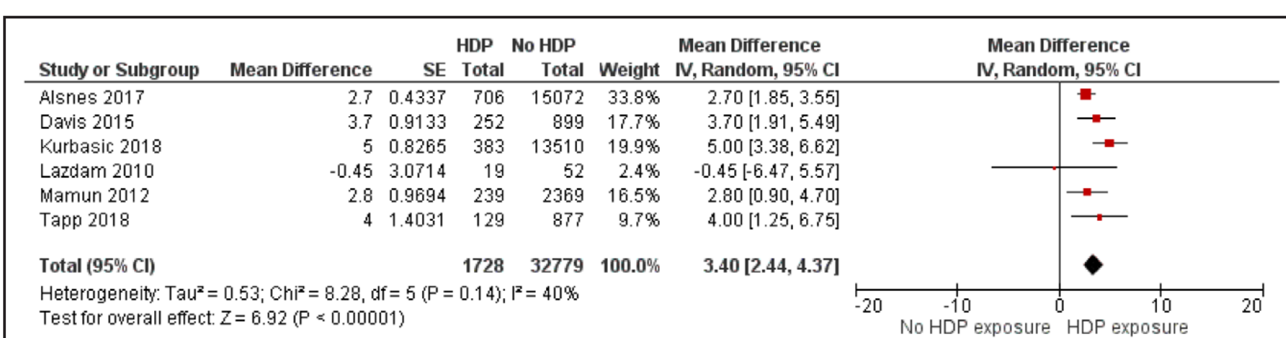

Figure 2. Forest plots displaying systolic blood pressure in patients exposed to hypertensive disorders of pregnancy (HDP) compared with no HDP.

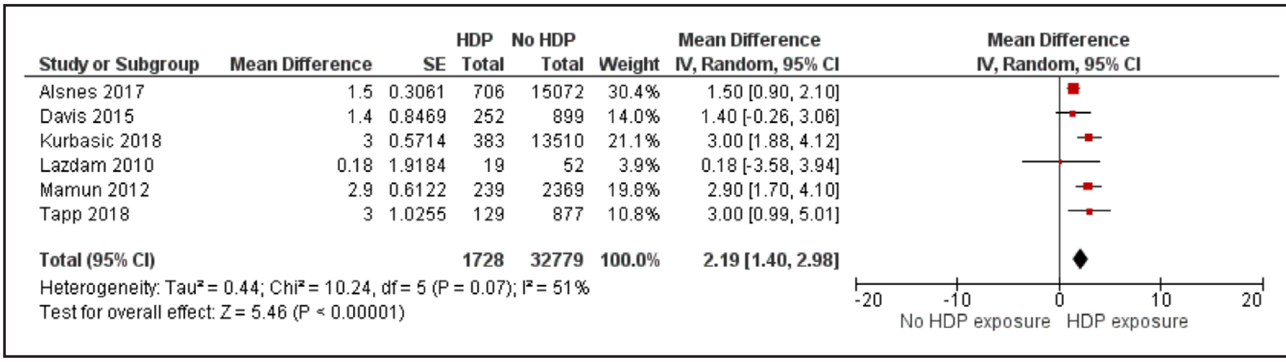

Figure 3. Forest plots displaying diastolic blood pressure in patients exposed to hypertensive disorders of pregnancy (HDP) compared with no HDP.

In additional analysis, the most adjusted ORs from each study were pooled. 41% increased odds of hypertension (OR, 1.41 [95% CI, 1.17–1.69], I^2=43%, P=0.0003; Figure S8). Pooled adjusted OR for maternal body mass index also showed increased odds of hypertension (OR, 3.15 [95% CI, 1.04–9.59], I^2=66%, P=0.04; Figure S9).

Primary Outcome: Heritability of Hypertensive Disorders Between Mothers and Daughters

Six studies (n=216, 793) reported positive associations between HDP in mothers and daughters.[5–8,39,40] Pooled analysis of unadjusted OR and RRs showed an 85% increase in odds of hypertension (OR, 1.85 [95% CI,1.54–2.22], P<0.0001; Figure S10). Heterogeneity was high (I^2=85%). Subgroup analysis of HDP subtype—preeclampsia—explained some of the heterogeneity and showed more than double the odds of preeclampsia in adult offspring (OR, 2.06 [95% CI, 1.70–2.49], I^2=64%, P<0.0001; Figure S11). Odds of preeclampsia in first born (OR, 2.21 [95% CI, 1.95–2.51], I^2=0%, P<0.0001) and birth order higher than 1 (OR, 2.11 [95% CI, 1.85–2.41], I^2=0%; P<0.0001) were similar (Figures S14 and S15).

Pooled adjusted OR from 2 studies showed a 72% increased odds of HDP in adult offspring exposed to HDP (OR, 1.72 [95% CI, 1.41–2.09], I^2=81%, P<0.0001; Figure 5). A subgroup analysis assessing the effect of HDP subtype—preeclampsia—was performed which explained most of the heterogeneity (OR, 1.90 [95% CI, 1.47–2.46], I^2=36%, P<0.0001; Figure S16).

In additional analysis, pooled adjusted OR for maternal age showed increased odds of hypertension (OR, 1.83 [95% CI, 1.51–2.23], I^2=0%, P<0.0001; Figure S17). Furthermore, the most adjusted ORs from each study were pooled. 73% increased odds of hypertension (OR, 1.73 [95% CI, 1.42–2.11]; P<0.0001) was seen and with moderate heterogeneity (I^2=81%; Figure S18).

Narrative Review of Studies Not Eligible for Pooling

HDP and Coronary Heart Disease

Qanitha et al 2017, reported higher odds of premature acute coronary syndrome (OR, 5.0 [95% CI, 1.0–24.7]; P=0.050) associated with HDP after adjusting for various confounders.[11] Conversely, Kajantie et al 2009[10] reported no significant association between gestational hypertension and coronary heart disease (hazard ratio, 1.0 [95% CI, 0.8–1.3]; P=0.8).

HDP and Hypertension

Dines et al 2023[35] reported a 50% increased risk of chronic hypertension in offspring exposed to HDP (hazard ratio, 1.50 [95% CI, 1.18–1.90]; P<0.001).

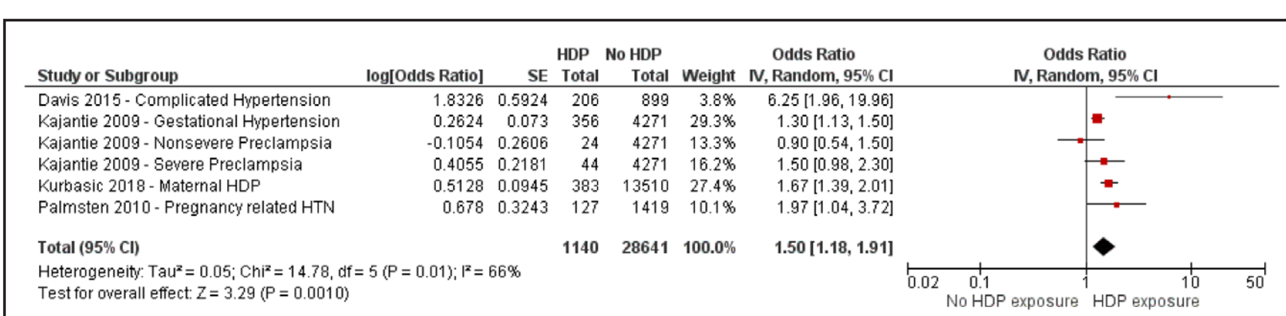

Figure 4. Forest plots displaying the odds of hypertension in patients exposed to hypertensive disorders of pregnancy (HDP) compared with no HDP adjusted for offspring sex.

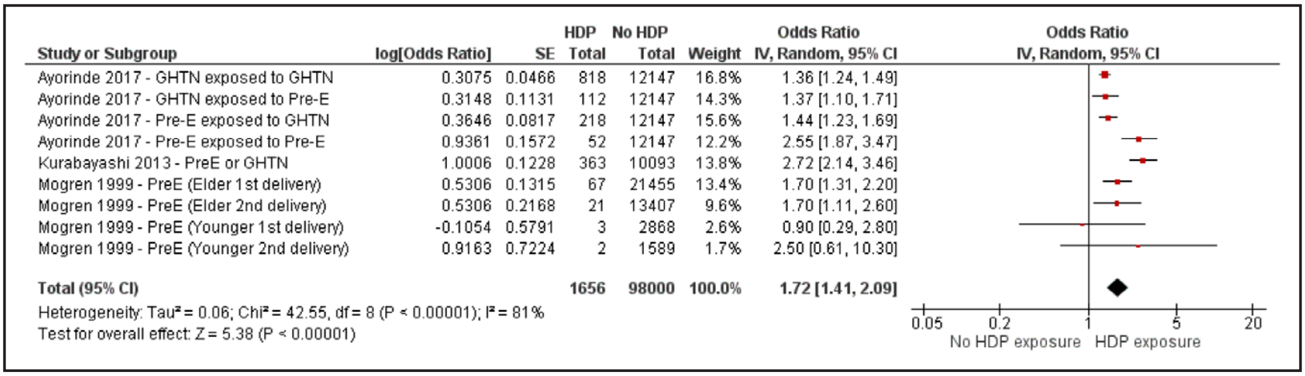

Figure 5. Forest plots displaying the odds of hypertensive disorders of pregnancy (HDP) in patients exposed to HDP compared with controls adjusted for offspring age.

HDP and Stroke

Kajantie et al 2009 found increased risk of thrombotic stroke linked to gestational hypertension (RR, 1.5 [95% CI, 1.1–2.2]; P=0.02) and greater risk of hemorrhagic stroke associated with nonsevere preeclampsia (RR, 3.2 [95% CI, 1.1–9.1]; P=0.03). Severe preeclampsia was associated with increased risk of stroke overall (RR, 2.2 [95% CI, 1.2–4.1]; P=0.01) and thrombotic stroke (RR, 2.5 [95% CI, 1.2–5.2]; P=0.01).[11]

Heritability of Hypertensive Disorders in Mothers and Daughters

Chesley et al, 1986 stated no association between eclamptic pregnancies and incidence of preeclampsia in daughters of eclamptic women.[42] Sutherland et al 1981[38] reported a 14% incidence of severe preeclampsia among mothers of preeclamptics compared with 3% incidence among mothers of controls. In addition, 16% of primigravid mothers of preeclamptics were found to have had severe preeclampsia, compared with only 3% incidence among mothers of controls. Finally, Arngrimsson et al 1990[41] concluded that the prevalence of preeclampsia and eclampsia in daughters was higher at 23% compared with 10% in controls (daughters-in-law).

DISCUSSION

Principal Findings

To the best of our knowledge, this is the first systematic review and meta-analysis analyzing the relationship between HDP exposure and cardiovascular outcomes in adult offspring, with an additional analysis on the heritability of HDP among female offspring. In adulthood, offspring exhibit higher SBP and DBP. Secondly, HDP exposure is associated with increased odds of hypertension and increased odds of HDP following a maternal history of the disorder. Furthermore, narrative synthesis found increased odds of premature coronary syndrome and stroke associated with HDP and severe preeclampsia. As per the Grading of Recommendations, Assessment, Development and Evaluation assessment, there is a high degree of certainty when assessing SBP following exposure to HDP. This provides valuable evidence for decision making in policies and clinical guidelines, as the true effect is close to the estimated effect. Nonetheless, a high degree of uncertainty was established for all other outcomes, necessitating increased research in these areas.

Comparison With Previous Literature and Interpretation of Findings

The increase in systolic and diastolic mean difference in BP is the highest observed from previous studies.[9] Increased BP in adult offspring exposed to HDP can have severe long-term cardiovascular consequences. Mendelian randomization analysis showed a 10 mm Hg increase in genetically proxied SBP was associated with a 49% increased risk of incident cardiovascular disease, a 50% increased risk of coronary artery disease, and a 44% increased risk of stroke.[45] In keeping with these findings, Qanitha et al 2017[11] reported pregnancy-related hypertension was associated with 5× higher odds of premature acute coronary syndrome occurring in those aged 55 years or less after adjusting for various confounders. However, the exposure (defined as high BP) was inaccurately measured via take-home questionnaires completed by the participant or a family member. Responder bias and differential misclassification may have led to over- or underestimation of the association. Furthermore, there were only a total of 9 cases. Kajantie et al 2009[10] reported no association between HDP and coronary heart disease in a cohort of 464 cases. Despite this, a 50% increased risk of thrombotic stroke was linked to gestational hypertension, greater than 3× increased risk of hemorrhagic stroke was associated with nonsevere preeclampsia, and severe preeclampsia was associated with greater than 2× increased risk of stroke overall (both thrombotic and hemorrhagic). In our review, only 2 out of 11 studies reported cardiovascular events.[10,11] The increases of 2 to 3 mm Hg in BP may be causing increases in cardiovascular risk that are too small to detect as significant cardiovascular events in early and middle adulthood. Hence, observational studies monitoring offspring into later adulthood are required.

Target organ damage may be an early marker that precedes hypertension.[31] Tapp et al 2018 found changes in retinal microvasculature and cardiac structure, including narrowed and longer retinal arteriolar diameters and increased left atrial volume indexed to body surface area. These changes may precede onset of hypertension and may have consequences on cardiovascular recovery and mortality among offspring exposed to HDP. A systematic review assessing offspring from childhood found changes to the structure and function of the heart, including ventricular mass, ejection fraction, and stroke volume in offspring exposed to HDP.[45] The following vascular changes were also noted: higher aortic intima-media thickness, decreased occipital and parietal vessel size, and a lower ratio of arteriolar-to-venular ratio. Future research using MRI to assess organ structure and function has been suggested, as well as cohort studies with various data timepoints in the same individual to assess the role of environmental and lifestyle factors.

Genetic influence of HDP is conflicting. Alsnes et al 2017[36] and Kurbasic et al 2018[32] found similar cardiovascular risk in siblings exposed and unexposed to HDP. In other words, siblings born to normotensive pregnancies also had increased cardiovascular risk. This suggests a familial predisposition that could be driven by genetic, environmental or behavioral factors. Following analysis of 122 733 cases for coronary artery disease and 34 217 cases for ischemic stroke, genetically predicted HDPs were linked to 24% and 27% increased odds of coronary artery disease and ischemic stroke, respectively.[47] Conversely, no association was found between maternal unweighted genetic scores and offspring cardiometabolic outcomes following assessment of 29 708 genotyped mother-offspring pairs.[23] Furthermore, epigenetic changes including DNA methylation, were found to be different in those exposed versus unexposed to preeclampsia.[48] Additional changes to microRNA expression have been observed in offspring exposed to HDP.[49] However, several miRNAs have been also shown to be advantageous in reducing cardiovascular risk depending on level of expression.[50] Therefore, although microRNA and epigenetics have potential use as biomarkers to predict cardiovascular risk in offspring, further research is required.

A graded association between severity of HDP and risk of hypertension was observed (Davis et al 2015). Similarly, a dose response effect with a higher risk of preeclampsia in daughters exposed to maternal preeclampsia compared with gestational hypertension was found.[6] Daughters with severe or early-onset preeclampsia were also 3× at greater odds of being born to preeclamptic pregnancy than normotensive pregnancy.[5] Unlike gestational hypertension, preeclampsia is characterized by widespread endothelial dysfunction causing hypertension, increased capillary permeability, thrombocytopenia, and disseminated intravascular coagulation.[51] Preeclampsia is also associated with placental dysfunction characterized by fetal growth restriction, oligohydramnios, and abnormal Doppler analysis of the umbilical or maternal uterine artery.[51] The Fetal Medicine Foundation (FMF) algorithm has been shown to double the detection rate of preterm preeclampsia and maybe indicated in daughters at higher risk of HDP. FMF involves the following measurements: uterine artery Doppler, placental biomarkers (pregnancy-associated plasma protein A (PAPP-A) and PlGF [protein placental growth factor]), maternal risk factors, and BP.[52,53] When used in conjunction with aspirin, the FMF algorithm was associated with a significant decrease in preterm preeclampsia. The FMF algorithm should be evaluated for routine clinical practice, especially in high-risk populations such as following offspring exposure to HDP.[52]

The effect of HDP exposure on hypertension and BP and the inheritability of HDP may have been overestimated due to insufficient adjustment for confounders. Yu et al 2022[9] found offspring aged >18 years had higher SBP and DBP than those aged <18 years. Age also resulted in some heterogeneity. We cannot exclude the confounding effects of age or body mass index in our meta-analysis. For odds of hypertension, 1 of 5 studies adjusted for age and 2 of 4 studies adjusted for body mass index.[10,29,32] When measuring the heritability of HDP, 4 out of 6 studies adjusted for age and 2 out of 6 studies for body mass index. One study adjusted for smoking and results showed a protective effect.[6] However, misclassification, loss of participants and incorrect adjustments have been shown to introduce bias when assessing smoking as a confounder.[54]

In addition, 7 studies used diagnostic codes for HDP exposure status and outcomes rather than standardized clinical measurements by a trained and blinded health care professional.[7,10,32,33,35,39,40] Outcomes such as hypertension were self-reported in 3 studies, introducing information bias through misclassification.[31,32,34,55] Despite variation of HDP risk in different ethnicities, there was a lack of studies from the African and Asian continents. These regions also have the highest prevalence of HDP.[56–58]

Strengths and Limitations

Our systematic review has highlighted important public health findings associated with HDP exposure and cardiovascular outcomes specifically in adult offspring. This is also the first meta-analysis investigating HDP among daughters exposed to maternal HDP. There are some limitations. HDP definitions varied between studies originating from 1990 to 2023. HDP subtypes were not investigated by all studies, and thus subgroup analysis was not possible. As a systematic review, publication bias cannot be excluded with certainty. We have tried to address this. The funnel plot was asymmetrical when assessing odds of HDP among daughters exposed to maternal HDP with high heterogeneity, indicating modest publication bias favoring a slightly large effect estimate. There is decreased certainty in the pooled estimate, as results maybe appearing slightly more positive due to overestimation of the true effect size. Confounders adjusted for in each study varied, and offspring age could not be adjusted for when pooling for odds of hypertension in offspring exposed to HDP. Future studies should consider age at onset of menopause. This could be an important confounder when investigating cardiovascular risk into later adulthood. Statistical testing such as Egger could not be conducted to assess publication bias. An assumption of the Egger test is a minimum of 10 pooled studies. However, our meta-analysis only included 9. Furthermore, we acknowledge the weakness of the DerSimonian-Laird method for pooling few numbers of studies.

Clinical Implications

The findings highlight the need for family-based screening and prevention strategies. Offspring exposed to HDP and those with a family history of HDP should receive yearly cardiometabolic screening from early adulthood, including BP monitoring, ECG monitoring, and calculation of QRISK score, ensuring timely diagnosis and preventative treatment. Lifestyle modification, including physical activity and healthy diet should be encouraged opportunistically during screening to target potential environmental and behavioral mechanisms. Early delivery remains the only treatment for preeclampsia, despite resulting in severe maternal and neonatal outcomes.[52] Thus, aggressive screening during the first trimester and prevention strategies, such as using the FMF algorithm and aspirin, should be considered in daughters exposed to HDP. Detection rates of preterm preeclampsia are as high as 75% with the FMF algorithm.[59]

Conclusions

HDP are associated with cardiovascular disease in adult offspring, including hypertension, acute coronary syndrome, and stroke. Female offspring exposed to HDP are at higher odds of inheriting HDP themselves. Future studies should investigate HDP subtypes and increase duration of follow-up into late adulthood. There should also be increased representation of different ethnicities in observational studies due to varying HDP risk. The detrimental effect of HDP on adulthood warrants increased monitoring, targeted prevention, and treatment strategies.

ARTICLE INFORMATION

Affiliations

Institute of Applied Health Sciences, School of Medicine, Medical Sciences and Nutrition, University of Aberdeen, United Kingdom (A.M.S., T.A.P., P.K.M.). University Hospitals of Leicester NHS Trust, Leicestershire, United Kingdom (R.C.). Department of Biostatistics and Health Informatics, Institute of Psychiatry, Psychology and Neuroscience. King's College London, United Kingdom (B.C.).

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Author Contributions

The corresponding author attests that all listed authors meet authorship criteria and that no others meeting the criteria have been omitted. SB conceptualized the research idea and supervised the initial phase of the project. A.M. Saji and R. Chandarana carried out database searching, screening, data extraction, and quality assessment. A.M. Saji analyzed the data under supervision of T.A. Pana, B. Carter, and P.K. Myint. A.M. Saji wrote the article. T.A. Pana, B. Carter, and P.K. Myint reviewed and edited the article. All authors read and approved the final article. Guarantor: P.K. Myint

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